

Professional Summary

Data Analyst with 3+ years of experience in analyzing large, complex datasets and developing scalable, data-driven solutions using Python, SQL, R, Snowflake, and Scala. Proficient in building automated ETL pipelines, optimizing data models, and creating interactive dashboards using Power BI, Tableau, and Excel to support decision-making and improve operational performance. Skilled in statistical analysis, predictive modeling, NLP, and advanced analytics using libraries such as Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Keras, and PyTorch. Experienced in collaborating with cross-functional teams to translate business requirements into analytical insights and automated reporting solutions. Strong hands-on expertise with AWS and GCP cloud services, distributed computing tools like PySpark and Hadoop, and handling sensitive data securely in compliance with governance and regulatory standards.

Technical Skills

Programming Languages: Python, SQL, R, Shell Scripting, Scala
Python Tools & Libraries: Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Keras, PyTorch, Matplotlib, Seaborn, SpaCy
Cloud Technologies: AWS (S3, Lambda, Glue, Redshift), GCP (BigQuery, Cloud Storage, Dataflow)
Data Visualization Tools: Power BI, Tableau, Excel (Pivot Tables, Power Query), Matplotlib, Seaborn
Data Engineering & ETL: ETL Pipelines, Data Cleaning, Data Validation, Automated SQL Jobs, Data Modeling, Data Warehousing, Workflow Automation
Healthcare Domain Expertise: Clinical Data, EHR/EMR Data, Patient Encounter Data, Claims & Billing Data, Provider Network Data, Member Eligibility Data, HEDIS Measures, Care Management Data, Quality Reporting (CMS, NCQA), HIPAA Compliance
Machine Learning & Advanced Analytics: Predictive Modeling, Statistical Analysis, NLP, Anomaly Detection, Trend Analysis, Cohort Analysis, Utilization Analysis, Correlation Analysis, Segmentation & Clustering, Time-Series Forecasting, Outlier Detection

Technical Skills

Data Analyst HCA Healthcare, USA	Nov 2024 - Present
- Analyzed large-scale clinical, patient encounter, EHR, and operational datasets using SQL, Python, and Snowflake to improve patient outcomes, care quality, and workflow efficiency. - Built automated ETL workflows and performed data quality validation using Python (Pandas, NumPy) to enhance clinical data accuracy and streamline reporting pipelines. - Designed and delivered Power BI dashboards for KPIs including hospital utilization, readmission rates, LOS, patient safety indicators, and throughput performance. - Developed optimized SQL queries and Snowflake data models to improve dashboard performance, reduce refresh delays, and support cross-department analytics needs. - Performed comparative analysis, variance analysis, and cohort analysis to identify trends, care gaps, and performance deviations across clinical departments. - Conducted anomaly detection, diagnostic analytics, and statistical deep-dives to uncover workflow bottlenecks and high-impact drivers for quality improvement initiatives. - Generated automated reports using SQL, Excel (Pivot Tables, Power Query), and Tableau to support patient safety reviews, regulatory audits, and operational performance monitoring.	

Data Analyst Persistent Systems, India	Sep 2020 - Nov 2022
- Analyzed large-scale healthcare claims, member eligibility, and provider data using SQL, Python, and ETL pipelines, improving data accuracy and operational reporting. - Built automated data cleaning and validation scripts using Python (Pandas, NumPy) to detect missing values, anomalies, coding errors, and duplicate claims, reducing manual processing efforts by 40%. - Developed interactive Power BI dashboards to track claims processing KPIs, denial trends, utilization metrics, and cost-per-member, enabling leadership to make data-driven decisions. - Created data models and optimized complex SQL queries, improving report refresh performance by 30% across multiple healthcare operations dashboards. - Collaborated with cross-functional teams including business SMEs, QA engineers, and data architects to translate requirements into analytical datasets and reporting solutions. - Performed root cause analysis on claim denials, billing discrepancies, and provider coding issues, delivering insights that reduced rework and improved first-pass adjudication rates. - Generated automated weekly/monthly analytical reports using Excel (VLOOKUP, Pivot Tables), SQL, and Power BI, supporting compliance, audit activities, and operational monitoring.	

Education

Master of Science (M.S.) in Computer Science and Engineering	Jan 2023 - Dec 2024
Florida Atlantic University, Boca Raton, Florida	

Projects

Sales Performance Analysis for a Retail Chain Tech Stack: Power BI, SQL, Excel, Python
- Analyzed multi-regional sales data to identify revenue trends, top-performing products, and underperforming stores. - Built interactive Power BI dashboards to track sales KPIs. - Provided insights that improved marketing efficiency and boosted business performance by 20%.
Healthcare Appointment Optimization System Tech Stack: Python, Power BI, SQL, Azure Synapse
- Developed predictive models to identify high-risk appointment cancellations and improve clinic scheduling. - Integrated no-show probability outputs into Power BI dashboards for real-time operational visibility. - Increased scheduling efficiency and patient attendance by 18% through data-driven automation.